

SignalFlow Traffic Signal Control Simulator

Project Planning and Functional Specification

This document defines the planned scope, interface, controls, logic, data handling, and implementation sequence for the SignalFlow web application. It is suitable for the planning and specification section of a project report.

Project item	Specification
Project type	Interactive web based traffic signal control simulator
Core purpose	Model a traffic signal as a state driven interface with manual control, timed automation, feedback, and stored preferences
Visual direction	A realistic city intersection at violet dusk, presented through synchronized background scenes
Final focus	Signal control, timing, state management, and user interface feedback

Document scope

The specification covers the features that belong to SignalFlow as established in the project plan. It does not add traffic or vehicle simulation, extra pages, sound, theme switching, or other unrelated enhancements.

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1 Project Definition

Project Title

SignalFlow Traffic Signal Control Simulator

Objective

SignalFlow transforms a basic traffic light button exercise into an interactive web application that controls a signal through a central state. The project demonstrates DOM manipulation, event handling, timers, conditional logic, objects and arrays, keyboard events, localStorage, and dynamic interface updates using vanilla JavaScript.

Concept

The simulator represents a traffic signal at a city intersection at violet dusk. A user may select a signal manually or start an automatic sequence. The current signal state is the source of truth: the background scene, current state panel, next state, countdown, activity log, statistics, and system status respond to that same state.

Application model

- User action or automatic timer
- Signal state
- Background scene and current state panel
- Countdown and next state
- Activity log and statistics

How SignalFlow Exceeds a Basic Traffic Light Exercise

Basic exercise	SignalFlow
Independent buttons directly change the visible light.	A centralized signal state drives all dependent interface updates.
Only a static red, yellow, or green result is shown.	Manual operation, timed automatic operation, pause and resume, presets, and emergency behavior are supported.
No operational history or measurement.	The activity log and live session statistics record what happens during the simulator session.
Simple light elements or colour changes.	Four matching city-scene images crossfade so the physical signal appears to change within one environment.
No persistence or secondary views.	Analytics, settings, keyboard options, and localStorage preferences complete the application scope.

2 Scope Boundaries

SignalFlow is a signal control simulator. Its purpose is to demonstrate a reliable state driven web interface. The project therefore remains focused on the signal, its timing, its visual feedback, and the information needed to operate it.

Included Scope

- Manual red, yellow, and green signal control from an OFF starting state.

- Automatic red to yellow to green to yellow to red cycling with countdown, pause, resume, and reset.
- Normal, Night, Peak Hour, and Emergency presets.
- Custom red, yellow, and green timing with validation and an Apply Settings action.
- An activity log, live statistics, keyboard shortcuts, analytics, settings, localStorage preferences, and responsive layout.

Explicitly Removed

Removed item	Reason it is outside this project
Traffic or vehicle simulation	The project does not simulate traffic flow, road conditions, or vehicle decisions.
Car movement and vehicle animation	These elements would shift the work away from signal control and state management.
Traffic simulation panels	The simulator does not require a separate traffic dashboard to meet its learning objective.
Unnecessary navigation pages or dashboard elements	The navigation remains limited to Simulator, Analytics, Settings, and live system status.

3 Visual Design and Four Image Approach

Violet Dusk Palette

The interface uses the Violet Dusk palette to match the evening city setting. Signal colours remain reserved for signal state feedback, rather than being used across the whole interface.

Colour	Role in the interface
#502D55	Deep background and primary dark surfaces
#935073	Cards, borders, and accent surfaces
#F6DBC0	Primary buttons and highlights
#F8F4E9	Main text and light surfaces
Red, yellow, green	Reserved for current signal state and related feedback

Synchronized Scene Images

The traffic light is part of the environment rather than a separate set of HTML circles. Four image files represent the same city intersection. Each image keeps the camera, road, buildings, sky, trees, signal location, signal scale, environmental lighting, and composition unchanged. Only the illuminated signal lamp changes.

State	Asset	Visible result
OFF	assets/signal-off.png	All traffic signal bulbs are off.
RED	assets/signal-red.png	Only the red lamp is illuminated.
YELLOW	assets/signal-yellow.png	Only the yellow lamp is illuminated.

State	Asset	Visible result
GREEN	assets/signal-green.png	Only the green lamp is illuminated.

The delivered image set uses the same 1672 by 941 pixel dimensions. Matching dimensions and composition prevent the scene from jumping when the signal changes.

Image Crossfade

Two stacked background layers are used for image switching. The next image is loaded into the hidden layer, then the opacity of the layers is exchanged. This produces a smooth crossfade from OFF to RED or between active signal states while preserving the impression of one continuous scene.

4 Technology Stack

The project uses browser technologies only. No framework is required because the purpose is to demonstrate the underlying JavaScript concepts directly.

Technology	Use in SignalFlow
HTML5	Semantic page structure, navigation, control groups, input fields, status cards, log, analytics, and settings views.
CSS3	Violet Dusk styling, responsive layout, buttons, cards, visual state feedback, image layer positioning, and crossfade transitions.
Vanilla JavaScript	Central state management, DOM updates, event listeners, timers, validation, statistics, activity logging, keyboard handling, and navigation.
Browser localStorage	Persists timing preferences and settings that the user chooses to save.
CSS transitions and animations	Supports the smooth scene crossfade and Night preset yellow blinking.

5 Interface Structure and Navigation

The interface is deliberately compact. The main simulator presents controls, the city scene, the current signal state, activity information, statistics, and shortcut help without adding unrelated dashboard panels.

Simulator Layout

Area	Contents and purpose
Navbar	SignalFlow identity, navigation actions, and the live system status.
Control panel	Mode selection, manual signal controls, automatic controls, presets, timing fields, and Apply Settings.
Scene area	The violet dusk city intersection rendered through the state based background image layers.
Current state panel	Current state, remaining countdown, next state, and paused or emergency feedback.
Information panels	Activity log, session statistics, keyboard shortcuts, analytics, and settings content.

Navbar Items

Item	Function
SignalFlow	Returns to the main Simulator view. It fulfils the role of a home action, so a separate Home page is not needed.
Simulator	Opens the primary operation view containing signal controls, timing, presets, the scene, and the current state.
Analytics	Opens the session data view using the statistics generated by the simulator, including signal usage bars.
Settings	Opens the preference view for remember timing, keyboard shortcut enable or disable, saving preferences, localStorage persistence, and project information.
System Status	A live indicator, not a separate page. It reports System Online, Auto Mode Running, Paused, or Emergency as appropriate.

6 Controls and Modes

All primary interactions are routed through the central signal state rather than performing isolated visual changes. Availability changes when the simulator enters automatic, paused, Night, or Emergency behaviour.

Mode Controls

Control	Function	Expected result
Manual mode	Places the simulator under direct user control.	RED, YELLOW, and GREEN controls are available. The user selects the required signal state.
Auto mode or Start Auto	Starts the timed state sequence.	The simulator enters AUTO and follows RED -> YELLOW -> GREEN -> YELLOW -> RED with a countdown for each phase.
Pause	Stops the active automatic countdown.	The current signal remains visible and the remaining seconds are preserved. System Status reports Paused.
Resume	Continues a paused automatic countdown.	The countdown continues from its preserved remaining value instead of restarting the phase.
Reset	Returns the simulation to its initial operational state.	Active timers stop, auto mode stops, the signal returns to OFF, the OFF image appears, countdown and temporary UI state clear, and manual mode becomes active. Session statistics remain available.

Manual Signal Controls

Control	Central action	System updates
RED	setSignal red	Shows signal-red.png, sets Current State to RED, identifies YELLOW as the next normal state, updates state feedback, records RED activated, and increments red activations.
YELLOW	setSignal yellow	Shows signal-yellow.png, sets Current State to YELLOW, sets the appropriate next normal state, updates feedback, records YELLOW activated, and increments yellow activations.

Control	Central action	System updates
GREEN	setSignal green	Shows signal-green.png, sets Current State to GREEN, identifies YELLOW as the next normal state, updates feedback, records GREEN activated, and increments green activations.

In manual mode, a selected signal updates the state immediately. A countdown is meaningful during automatic operation; manual selections do not create a second automatic timer.

7 Signal Logic and Countdown

Central State Machine

SignalFlow uses a state machine to control valid automatic transitions. The automatic cycle includes two yellow phases: one after red before green, and one after green before red. The phase position is retained so the simulator can choose the correct next state.

Current phase	Next automatic phase	Purpose
OFF	RED when Auto starts	The initial safe and inactive state.
RED	YELLOW before green	Begins an automatic cycle and prepares the change to green.
YELLOW after red	GREEN	Transition from stop to go.
GREEN	YELLOW before red	Prepares the return to stop.
YELLOW after green	RED	Completes the transition and returns to the next cycle.

Automatic Countdown

When Auto begins, the signal starts at RED and receives the configured red duration. The remaining time counts down once per second. At zero, the state machine selects the next phase, applies the relevant signal image and timing value, then restarts the countdown for that phase. Reaching RED after the green-side yellow phase increments the completed cycle count.

Example normal operation

- RED for 10 seconds
- YELLOW for 3 seconds
- GREEN for 8 seconds
- YELLOW for 3 seconds
- Return to RED and begin the next cycle

Current State Panel

The panel presents the active signal, the remaining time during Auto, and the next state. When paused, it shows PAUSED with the preserved number of seconds remaining. When Emergency is active, it shows the red signal and the emergency state rather than a running transition.

8 Presets and Custom Timing

Operating Presets

Preset	Behaviour
Normal	Standard automatic timing: red 10 seconds, yellow 3 seconds, and green 8 seconds.
Night	Uses a blinking yellow signal. Other lights remain off while the yellow state repeatedly blinks.
Peak Hour	Uses a longer green phase for the simulation. Established example timing is red 8 seconds, yellow 3 seconds, and green 15 seconds.
Emergency	Stops normal automatic operation, forces RED, reports Emergency, and disables normal controls while the emergency condition is active.

Custom Timing

The control panel provides editable fields for Red, Yellow, and Green duration in seconds, followed by Apply Settings. The user can choose timing values appropriate to the simulator session without changing the underlying state model.

Input or action	Function
Red timing	Sets the configured duration used for the red phase during future automatic operation.
Yellow timing	Sets the configured duration used for both yellow transition phases during future automatic operation.
Green timing	Sets the configured duration used for the green phase during future automatic operation.
Apply Settings	Reads the three values, validates them, stores valid values in the timing configuration, updates the interface, and makes them available to Auto mode.

Validation Rules

Each timing value must be a positive numeric duration. Empty values, text such as abc, zero, and negative values are invalid. The system shows a validation message and does not apply any invalid timing set. If settings are changed while Auto is running, the new values take effect from the next relevant phase so the active phase is not unexpectedly restarted.

9 User Feedback and Supporting Pages

Activity Log

The activity log records meaningful actions with the time and relevant state. It gives the user a short operational history without adding a separate traffic panel.

Event type	Example log message
Manual signal selection	RED activated, YELLOW activated, or GREEN activated
Automatic operation	Auto cycle started, Auto cycle paused, or Auto cycle resumed

Event type	Example log message
Configuration	Timing updated
Special operation	Night mode enabled or Emergency mode enabled
Reset	System reset

Session Statistics

Statistic	Meaning
Cycles	Number of automatic cycles completed.
Changes	Number of signal state changes made during the session.
Red activations	Count of times RED becomes the active signal.
Yellow activations	Count of times YELLOW becomes the active signal.
Green activations	Count of times GREEN becomes the active signal.
Runtime	Elapsed session running time displayed in a readable time format.

Keyboard Shortcuts

Key	Function
R	Set RED in the allowed operating state.
Y	Set YELLOW in the allowed operating state.
G	Set GREEN in the allowed operating state.
A	Toggle Auto mode or stop automatic operation.
Space	Pause or resume Auto mode.
E	Activate Emergency behaviour.
Esc	Reset the simulator.

Unknown keyboard keys do not change the simulator. Keyboard shortcuts are ignored when the settings preference disables them.

Analytics

The Analytics view reads the same session statistics shown in the simulator. It presents totals for cycles, state activations, and runtime, plus signal usage bars based on actual red, yellow, and green activations. Analytics does not invent a second data set; it visualizes the simulator session data.

Settings and localStorage

The Settings view includes the Remember Timing preference, keyboard shortcut enable or disable, a Save Preferences action, localStorage persistence, and project information. When timing persistence is enabled and

preferences are saved, the browser stores the chosen timing and relevant settings. On a later load, SignalFlow reads those saved values before presenting the simulator in its OFF and Manual initial state.

10 Data Architecture and Main Functions

The application keeps related data in central objects instead of scattering independent variables across event handlers. The signal state remains the source of truth for all visible and recorded behaviour.

Core Data Groups

Data group	Representative fields	Purpose
signalState	current, mode, next, remainingTime, isPaused, isEmergency, cycleCount	Describes the active signal and operational condition.
timings	red, yellow, green	Stores configured phase durations in seconds.
statistics	cyclesCompleted, changes, redActivations, yellowActivations, greenActivations, runtime	Provides the data for the statistics panel and Analytics view.
activityLog	action, time, state	Stores chronological user and system events for display.
preferences	rememberTiming, keyboardShortcutsEnabled	Stores user choices that may be saved through localStorage.

Main Function Responsibilities

Function	Responsibility
setSignal	Applies an allowed signal state and coordinates image, state text, next state, activity logging, statistics, and visual indicators.
startAutoMode and stopAutoMode	Start and stop the timed automatic sequence while ensuring that only one automatic timer exists.
pauseAutoMode and resumeAutoMode	Preserve and continue the remaining countdown time for an automatic phase.
startCountdown, stopCountdown, and updateCountdown	Manage the once-per-second countdown and request the next state when time reaches zero.
resetSystem	Stops active timers, returns the simulation to OFF and Manual, and clears temporary operation state without discarding session statistics.
applySettings	Reads timing values, validates them, updates timing configuration, and refreshes the relevant interface.
toggleEmergency and toggleNightMode	Apply the special Emergency or Night operating rules.
updateStatus and updateStatistics	Refresh the System Status, state panel, simulator statistics, and Analytics display from central data.
addActivity	Creates a timestamped activity record for the log.
handleKeyboard	Maps allowed key presses to the same actions as the visible controls.
saveSettings and loadSettings	Write eligible preferences to localStorage and restore them on page load.

11 User Flows and Edge Cases

Primary User Flows

Flow	Expected sequence
Page load	Load saved preferences if available -> show OFF scene -> set Manual mode -> display System Online.
Manual RED	User selects RED -> setSignal red -> crossfade to red scene -> update Current State and next state -> log event -> increment red activation.
Start Auto	User starts Auto -> set mode to AUTO -> set RED -> start red countdown -> transition through both yellow phases and green -> repeat.
Pause and resume	User pauses during Auto -> stop countdown without changing signal -> show Paused -> user resumes -> continue from preserved seconds.
Reset	User resets -> stop timers and Auto -> return to OFF image and Manual -> clear current temporary operation state.
Emergency	User selects Emergency or presses E -> stop normal operation -> force RED -> show Emergency -> disable normal controls while active.

Edge Cases and Expected Handling

Situation	Expected handling
Auto is selected repeatedly	Prevent duplicate timer instances. Only one automatic countdown may run at a time.
Pause is requested in Manual mode	Do not alter the signal or create a timer. The control may be unavailable or report that Auto mode is required.
Timing is changed during Auto	Validate the submitted values and use valid new timing from the next relevant phase rather than restarting the current phase.
Emergency begins during Auto	Stop the automatic timer, force RED, mark the system as Emergency, and restrict normal controls.
Invalid timing input	Show a validation message and retain the previously valid timing configuration.
Unsupported key is pressed	Ignore the key without changing the simulator.
Image state changes	Use the two-layer crossfade so the consistent city scene does not appear to jump.

12 Responsive Design and Folder Structure

Responsive Behaviour

The layout must remain usable without horizontal scrolling. Content is reorganized by screen size while retaining the same control and information hierarchy.

Viewport	Layout approach
Desktop	Three main regions: control panel, city scene and current state, and information panels.
Tablet	Vertical sequence: controls, scene, then information panels.

Viewport	Layout approach
Mobile	Header, scene, current state, controls, statistics, activity log, and keyboard shortcut help in a single accessible column.

Final Project Folder Structure

The delivered project keeps the application files together and stores the synchronized scene assets in one asset directory.

```
SignalFlow_Final/
  index.html
  style.css
  script.js
  README.md
  assets/
    signal-off.png
    signal-red.png
    signal-yellow.png
    signal-green.png
```

index.html defines the application content, style.css contains the visual system and responsive behaviour, script.js contains the signal logic and interface updates, and assets contains the four matching scene images. README.md is supporting repository documentation and is not required for the website to run.

13 Development Phases and Feature Map

Development Phases

Phase	Work completed in the phase
1 Foundation	Create HTML structure, Violet Dusk styling, responsive layout, navbar, the OFF state, and the four image layer system.
2 Core controls	Implement the central signal state, manual RED, YELLOW, and GREEN controls, current state feedback, image crossfade, Reset, and activity events.
3 Automatic operation	Add the automatic phase sequence, one managed countdown, pause and resume, cycle counting, and status updates.
4 Operating options	Add Normal, Night, Peak Hour, and Emergency behaviours plus custom timing and validation.
5 Supporting information	Add statistics, activity log, keyboard shortcuts, Analytics, Settings, localStorage preference saving, and project information.
6 Verification	Check that each planned control has an event handler, asset paths and dimensions match, responsive rules are present, special modes preserve constraints, and traffic simulation remains absent.

Feature Map

Feature area	Key elements	Purpose
Signal control	OFF, RED, YELLOW, GREEN, Manual, Auto	Select or automate a defined traffic signal state.
Timing	Countdown, Pause, Resume, Reset, custom	Manage a predictable automatic cycle and recover to a

Feature area	Key elements	Purpose
	durations	safe initial state.
Special operation	Normal, Night, Peak Hour, Emergency	Provide distinct operating behaviours without changing the core signal model.
Feedback	Scene crossfade, Current State, System Status, Activity Log	Make the current condition and each change visible to the user.
Measurement	Cycles, Changes, activations, Runtime, Analytics	Record and present real session data produced by the simulator.
Preferences	Remember Timing, keyboard enable or disable, Save Preferences, localStorage	Allow eligible user choices to persist between browser loads.

14 Final Scope and Project Distinction

SignalFlow is a complete traffic signal control simulator for the browser. Its final scope is a visual, state driven application with manual operation, controlled automatic cycling, time management, special presets, custom timing, live feedback, session information, keyboard access, responsive design, and saved preferences.

The project differs from a basic traffic light exercise because it does not treat each button as a separate colour change. The central signal state coordinates the city scene, countdown, next state, log, statistics, system status, analytics, and settings. The four synchronized image assets make the signal look integrated into one dusk intersection rather than an isolated set of coloured circles.

The project deliberately excludes vehicle movement and traffic simulation. That boundary keeps the work aligned with its intended learning outcomes: JavaScript state management, timing logic, event handling, persistent preferences, responsive interface design, and clear user feedback.